



5137 - Weighing the Intermediate Mass Black Hole in Omega Centauri

Cycle: 3, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec Primary				
	1	NIRSpec Target	NIRSpec IFU Spectroscopy	(1) OmegaCenCenter
	2	Background	NIRSpec IFU Spectroscopy	(2) OmegaCenBackground

ABSTRACT

We propose for deep NIRSpec IFU observations to follow up the detection of an intermediate mass black hole (IMBH) based on proper motion measurements of 7 fast-moving stars in the central 3" of Omega Centauri. These stars place a firm lower limit of 8000 solar masses on the black hole. This unique black hole is simultaneously (1) the most robust detection of any IMBH, (2) the nearest massive black hole, and (3) only the second with a resolved system of orbiting stars. Only with JWST NIRSpec IFU can we simultaneously measure the line-of-sight velocities of the stars and make the deepest available search for the expected accretion emission from the black hole. We will obtain line-of-sight velocities for ~70 stars, including the first measurements for all four fast moving stars within the central 1". In addition we will identify 3.1 ± 1.7 additional fast moving stars. These measurements will provide 3D velocity information crucial for the precise measurement of the IMBH mass based on the orbits of the stars. The proposed observations will also be sensitive to black hole accretion ~100x fainter than the upper limits placed by deep X-ray observations. Detection of any accretion source would identify the exact location of the black hole and reveal a black hole accreting orders of magnitude more faintly than any known.

OBSERVING DESCRIPTION

We propose a deep dithered set of observations in the center of Omega Centauri using the NIRSpec IFU with the G235H grating to obtain spectra from 1.66-3.17 microns. This data is centered on the Anderson & van der Marel (2010) position of Omega Centauri, around which proper motion measurements have revealed 4 stars moving faster than the escape velocity at $r < 1''$ and 7 stars at $r < 3''$. The focus of this proposal is measuring line-of-sight velocities to a precision of ~5 km/s for stars within the central region and searching for a faint accretion source with an expected red spectrum.

The data will be taken with NRSIRS2 readout with 8 groups/integration and 5 integrations per dither, with 12 dithers. This strategy optimizes detection of faint objects while not saturating even due to bright stars within the field. We use a medium dither pattern to enable our single background pointing to be used to make an independent background measurement for each pixel. The dithers will also provide excellent sub-pixel sampling which will be important for recovering the undersampled spatial PSF with the NIRSpec IFU.

Proposal 5137 - Targets - Weighing the Intermediate Mass Black Hole in Omega Centauri

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	OmegaCenCenter	RA: 13 26 47.2400 (201.6968333d) Dec: -47 28 46.45 (-47.47957d) Equinox: J2000	Proper Motion RA: -3.21 mas/yr Proper Motion Dec: -6.24 mas/yr Epoch of Position: 2000	
	Comments: Category=Stellar Cluster Description=[Globular star clusters] Extended=YES				
Fixed Targets	(2)	OmegaCenBackground	RA: 13 25 59.1000 (201.4962500d) Dec: -47 45 19.20 (-47.75533d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000	
	Comments: Empty region from F606W image and 2MASS/WISE imaging; update May 23rd 2025 to move away from FG star. Category=Stellar Cluster Description=[Globular star clusters] Extended=YES				

Proposal 5137 - Observation 1 - Weighing the Intermediate Mass Black Hole in Omega Centauri

Observation	Proposal 5137, Observation 1: NIRSpec Target											Fri May 23 21:00:10 GMT 2025	
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	OmegaCenCenter	RA: 13 26 47.2400 (201.6968333d) Dec: -47 28 46.45 (-47.47957d) Equinox: J2000				Proper Motion RA: -3.21 mas/yr Proper Motion Dec: -6.24 mas/yr Epoch of Position: 2000						
Template	<i>Comments:</i> Category=Stellar Cluster Description=[Globular star clusters] Extended=YES												
	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		MEDIUM		1		12					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G235H/F170LP	NRSIRS2	8	5	false	true	NONE	12	60	35888.669	171962	
	2	G235H/F170LP	NRSIRS2	8	5	true	true	NONE	12	60	35888.669		
Special Requirements	Group Observations 1, 2, Non-interruptible												

Proposal 5137 - Observation 2 - Weighing the Intermediate Mass Black Hole in Omega Centauri

Observation	Proposal 5137, Observation 2: Background											Fri May 23 21:00:10 GMT 2025	
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(2)	OmegaCenBackground	RA: 13 25 59.1000 (201.4962500d) Dec: -47 45 19.20 (-47.75533d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000						
Template	Comments: Empty region from F606W image and 2MASS/WISE imaging; update May 23rd 2025 to move away from FG star. Category=Stellar Cluster Description=[Globular star clusters] Extended=YES												
	TA Method						HFF Readout Mode						
Dithers	NONE						false						
	#	Dither Type		Size		Starting Point		Number of Points		Points			
Spectral Elements	1	NONE											
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
Special Requirements	1	G235H/F170LP	NRSIRS2	8	5	false	false	NONE	1	5	2990.722		
	Group Observations 1, 2, Non-interruptible												